



Republic of the Philippines

Department of Environment and Natural Resources

MINES AND GEOSCIENCES BUREAU

North Avenue, Diliman, Quezon City, Philippines

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LANDSLIDE AND FLOOD THREAT ADVISORIES

To: **Brgy. Chairperson**
Brgy. Marikina Heights
City of Marikina
Metropolitan Manila

Date: Oct 9, 2024

Dear Sir/Madam:

Please be advised that the Geohazards Mapping and Assessment Team (GMAT) of the Mines and Geosciences Bureau (MGB) has conducted **landslide and flood assessment** for the updating of 1:10,000 Scale Geohazard Map in your barangay. The following are the results and recommendations following the assessment:

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Flood Susceptibility Rating	Geohazards Assessment
Barangay Hall	N 14° 39' 04.7" E 121° 06' 56.9"	Moderate	Brgy. Marikina Heights is situated on a generally gently sloping terrain. No notable flooding was noted with most events attributed to rainfall runoff and/or topographic lows but these does not reach notable heights. Significant flooding, however, was reported on areas near creeks.
Champagnat	N 14° 39' 07.5" E 121° 06' 51.1"	Moderate	
Valenzuela	N 14° 39' 09.9" E 121° 06' 46.8"	Moderate	Marikina National High School and its vicinity was inundated with almost 2 m high flood waters during typhoon Ondoy. Source of the floodwaters may be a tributary of the Marikina River – a covered creek running under the parking lot. The pavement in several portions of the said lot is showing (at the time of the assessment) vertical displacement of as much as 1 cm. Several portions of the parking lot had been cordoned off for safety reasons.
Marikina Heights National Highschool	N 14° 38' 58.7" E 121° 07' 05.9"	High to Very High (River Erosion)	
Fortune Village I	N 14° 39' 12.3" E 121° 07' 22.5"	Moderate	Another area where notable flooding was reported was in a residential area called 'Valenzuela.' Significant flooding was reported during Typhoon Ondoy when the nearby creek/channel overflowed. Parallel cracks were observed along the road perpendicular to the creek. Residents reported that the cracks started appearing 1-2 years ago from the time of the assessment. The road is also sinking towards the river with some vertical replacement. This may be a sign of riverbank erosion. Flood mitigation along the river was constructed after typhoon Ondoy.
Unnamed creek / Apitong Street	N 14° 39' 12.5" E 121° 07' 21.1"	Moderate	
Remarks / Recommendations:			

- ✓ **Implement pre-emptive evacuation** by the City Disaster Risk Reduction Management Office (CDRRMO) and Barangay Disaster Risk Reduction Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons.
- ✓ **Construction and/or improvement of drainage system.** Existing channels should be broadened and deepened to ensure smooth drainage of floodwaters/runoff. Avoid improper waste disposal on channels as it may cause clogging and obstruction of water flow. Design of canals should also consider obstructions by buildings/infrastructures.
- ✓ **Strengthen and improve** flood warning system with corresponding warning levels for pre-emptive evacuation purposes
- ✓ Adhere to the warnings given by proper authorities prior to the landfall or passing of typhoon and other prevailing weather disturbances. This may include CDRRMO, Office of the Civil Defense, and other related agencies.
- ✓ **Construction of appropriate and well-studied engineering measures** (e.g. dike or flood wall etc.) should be implemented to prevent overtopping of floodwaters and the fast rate of erosion along rivers/creeks. Make sure that the structure follows engineering standards and has structural integrity. However, even though there are engineering measures, communities near the river/creeks must remain vigilant and not complacent during strong typhoons and inclement weather.
- ✓ The barangay should have **constant communication with nearby/upper barangays** during typhoon and monsoon season. This is to gather information from their counterparts in terms of the flood situation in their areas. Usually, floodwaters coming from nearby/upper barangays are the main cause of flooding within the barangay area. This may serve as preparation for evacuation of their constituents, especially those near rivers/creeks/canals.
- ✓ The CDRRMO, in coordination with the BDRRMO, should conduct information and education campaigns on flood hazards to increase public awareness, especially to identified flood-prone communities.

Purok / Sitio / Area of Concern	GPS Coordinates (Coordinate System: Luzon 1911)	Landslide Susceptibility Rating	Geohazards Assessment
Empress Village	N 14° 39' 05.0" E 121° 07' 24.4"	High	In Empress Village, a hilltop area can be observed. This area was identified by local officials as one of the possible landslide prone areas in the barangay. Accordingly, the its steep and exposed slope makes it susceptible to such. No outcrop was observed as the area was highly vegetated with a number of infrastructures established.

Remarks / Recommendations: ✓ Observe presence of mass movement (e.g. landslides, tension cracks) within the identified critical area/steep slopes and report to the local authorities and/or the MGB. ✓ Observe for saturated ground and/or displaced roads surfaces and report to the MGB/local authorities. ✓ Maintenance and/or implementation of appropriate slope stabilization (or slope protection/engineering intervention) measures along identified critical areas and roadcuts are highly recommended. ✓ Avoid settlement along the footslope, edge of the cliff/ravine, steep slope and ridge of hill/mountain. ✓ Barangay Disaster Risk Reduction Management Team must implement monitoring and pre-emptive evacuation during monsoon and typhoon season where extreme heavy rainfall occurs. ✓ Activate Barangay Disaster Risk Reduction Management Team in cooperation with the City Disaster Risk Reduction Management Team during monsoon and typhoon season. ✓ The CDRRMO together with the BDRRMO should conduct information and education awareness on this particular type of geohazard, particularly on landslide hazard to the landslide-prone communities.			

Prepared by:


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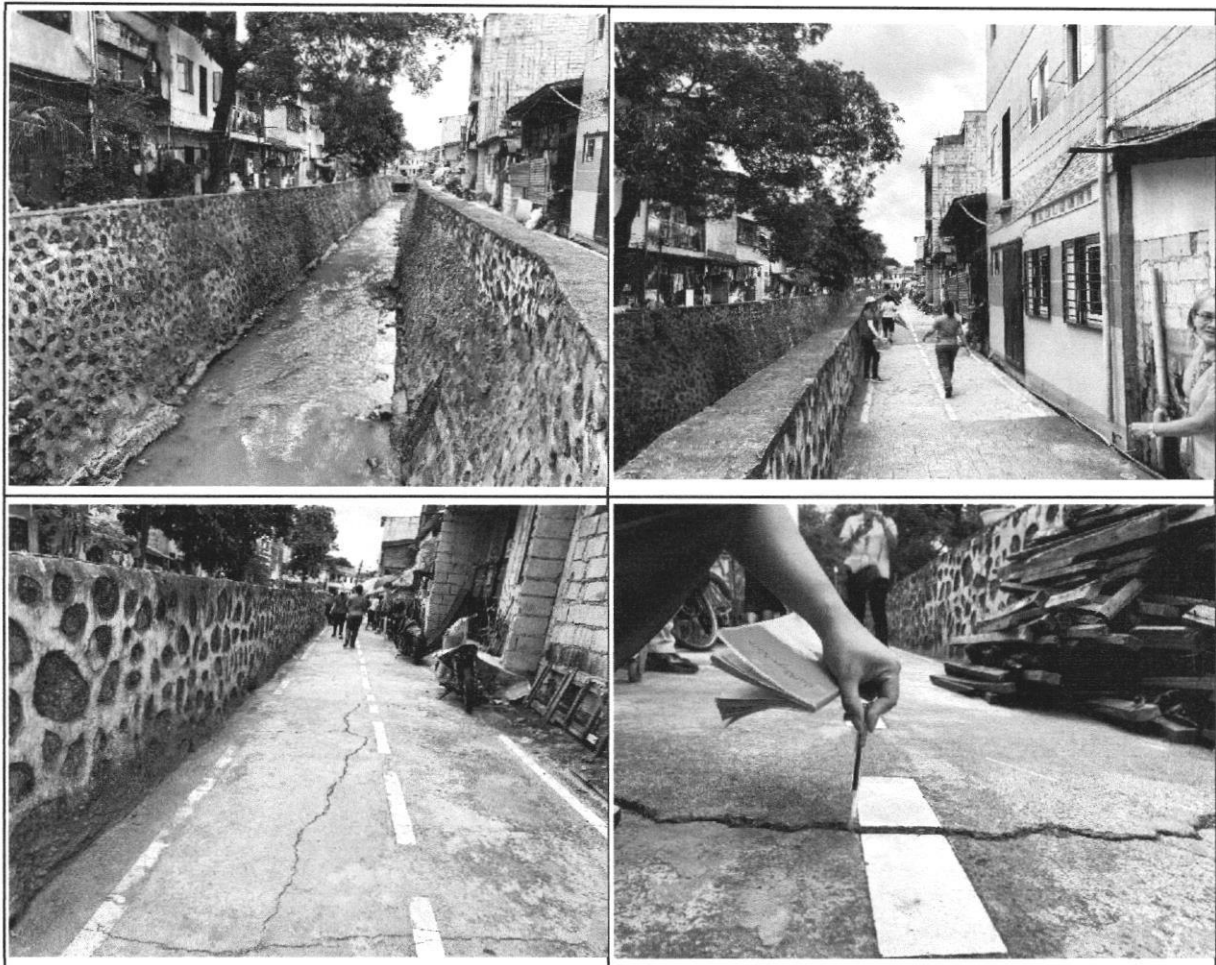

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Designation:

Office:

PHOTOS



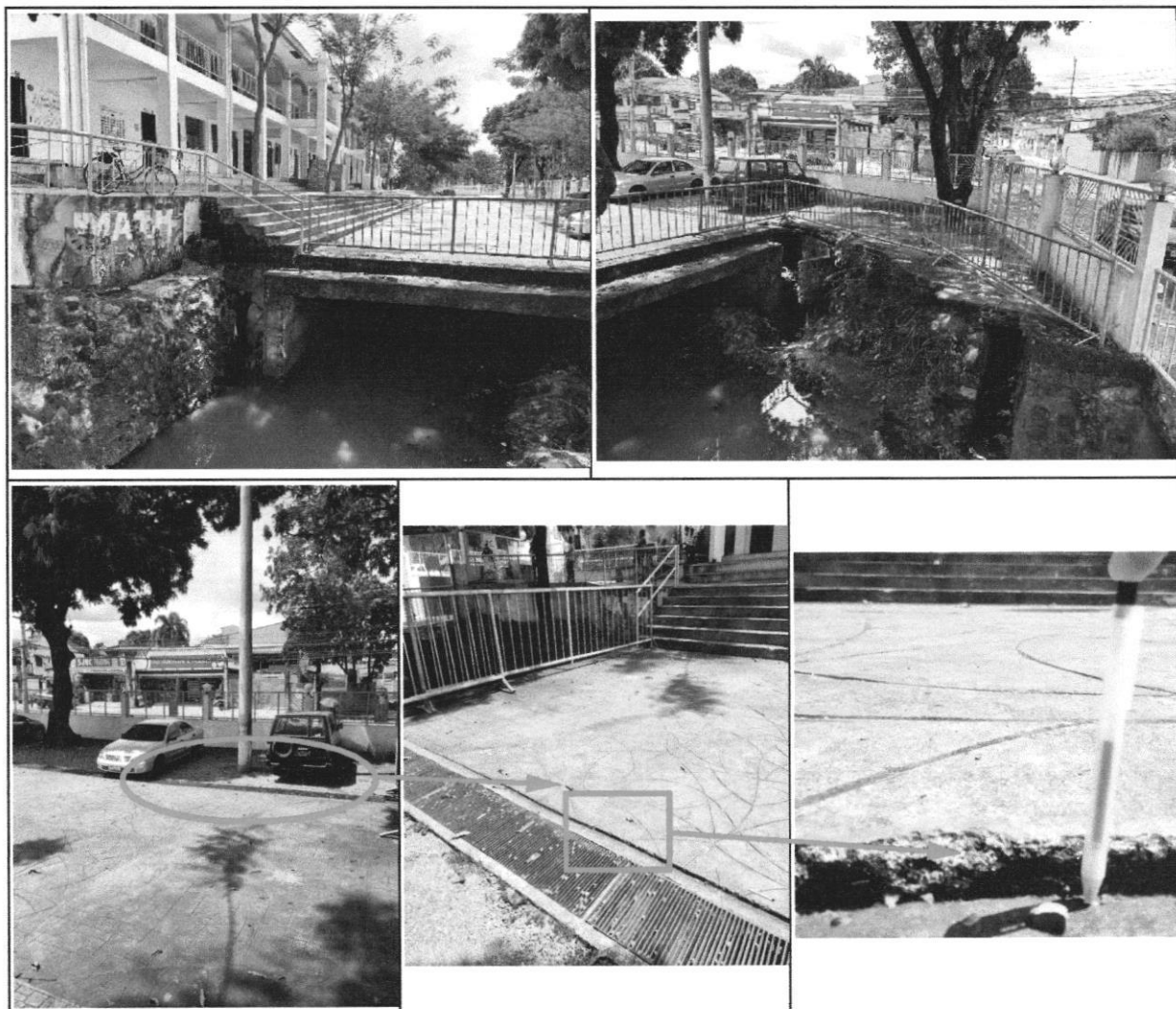
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Recommendations:

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Management Office (BDRRMO) during monsoon (heavy rainfall) and typhoon seasons.

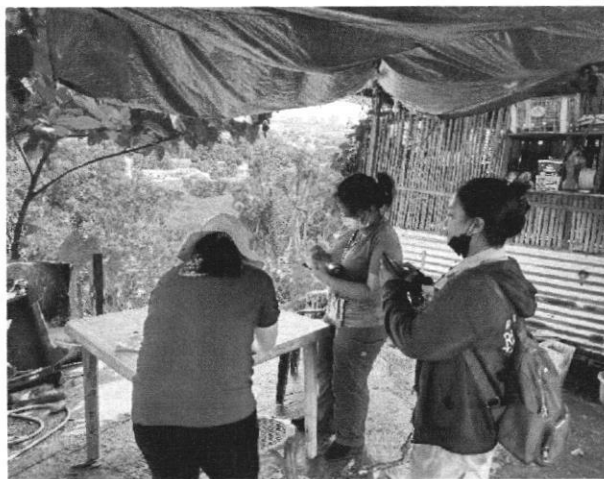
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View atop the hill in Empress Village. Possible erosion from surface runoff was observed. GPS coordinates: N 14° 39' 05.0", E 121° 07' 24.4")

Recommendations:

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